

Math 170S - Homework 7 - due March 5

Problem 1. Consider $X_1, \dots, X_n$ coming from the distribution $N(\mu, 100)$. Let $n = 20$. We test the hypothesis $H_0 : \mu = 60$ against $H_1 : \mu = 50$.

- a) For the critical region $\{\bar{X} \leq 57\}$ what are Type I and Type II errors?
- b) Without additional computations, how do the Type I and Type II errors change if we consider the critical region $\{\bar{X} \leq 54\}$ instead?
- c) Now, if we want Type I error to be 1%, what decision rule (critical region) should be? What will be Type II error then?
- d) Now we can change the sample size n . What is optimal n so that both Type I error is at most 1% and Type II is at most 3%?

Problem 2. Consider the distributions $N(\mu_X, 400)$ and $N(\mu_Y, 225)$. Let $\theta = \mu_X - \mu_Y$. Let $\bar{x}$ and $\bar{y}$ denote the observed means of two independent random samples, each of size n , from the respective distributions. Say we reject $H_0 : \theta = 0$ and accept $H_1 : \theta > 0$ if $\bar{x} - \bar{y} \geq c$. Let $K(\theta)$ be the power function of the test. Find n and c , so that $K(0) = 0.05$ and $K(10) = 0.90$, approximately.

Hint: First, you need to compute power function for testing the hypothesis about the difference of two means (by the definition of power function). Note that it is different from the formula we derived in class for testing the hypothesis about one mean.

Problem 3. 8.5-2 in the red book/8.6-2 in the blue book (it is the same problem in the section about power function)

Problem 4. 8.6-1 in the red book/8.7-1 in the blue book (from the section about best critical regions)

Problem 5. 8.6-3 in the red book/8.7-3 in the blue book (from the section about best critical regions)

Problem 6. 8.6-6 in the red book/8.7-6 in the blue book (from the section about best critical regions)